Name : Pallavi Rekhawar

Batch : DL3

Lab Assignment 3- Yolo Model

Part A: Object Detection

- **Train YOLO Nano / Small model**
- **Dataset: Roboflow detection dataset**

Screen Recording

https://drive.google.com/file/d/1HS_GuApNTbVYi8xiZUYyGz9ySlIdZ1Fv/view?usp=sharing

1. Title

Object Detection using YOLOv8 (Ultralytics Framework)

2. Objective

To design and implement an object detection system using YOLOv8 model to detect and classify objects from a custom dataset with high accuracy.

3. Tools & Technologies Used

- Python 3.11
- Ultralytics YOLOv8
- OpenCV
- Roboflow Dataset
- CMD

4. Dataset Description

- Source: Roboflow fruit dataset : https://universe.roboflow.com/ratul/fruit-detection-8ohhe
- Format: YOLO format
- Classes:
    - Apple-G1
    - Apple-G2

5. Model Used

- YOLOv8n (Nano model)
- Pretrained weights: yolov8n.pt
- Fine-tuned on custom dataset

6. Training Command Used

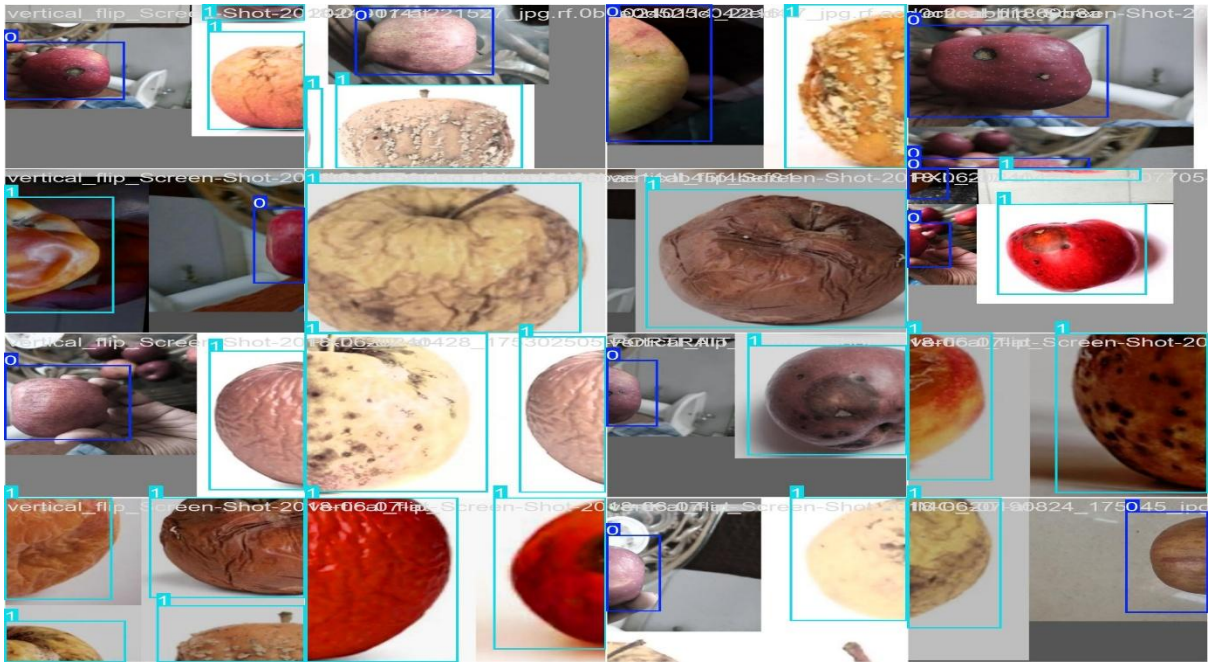
yolo detect train model=yolov8n.pt data=detection/data.yaml epochs=15 imgsz=640

7. Output Location

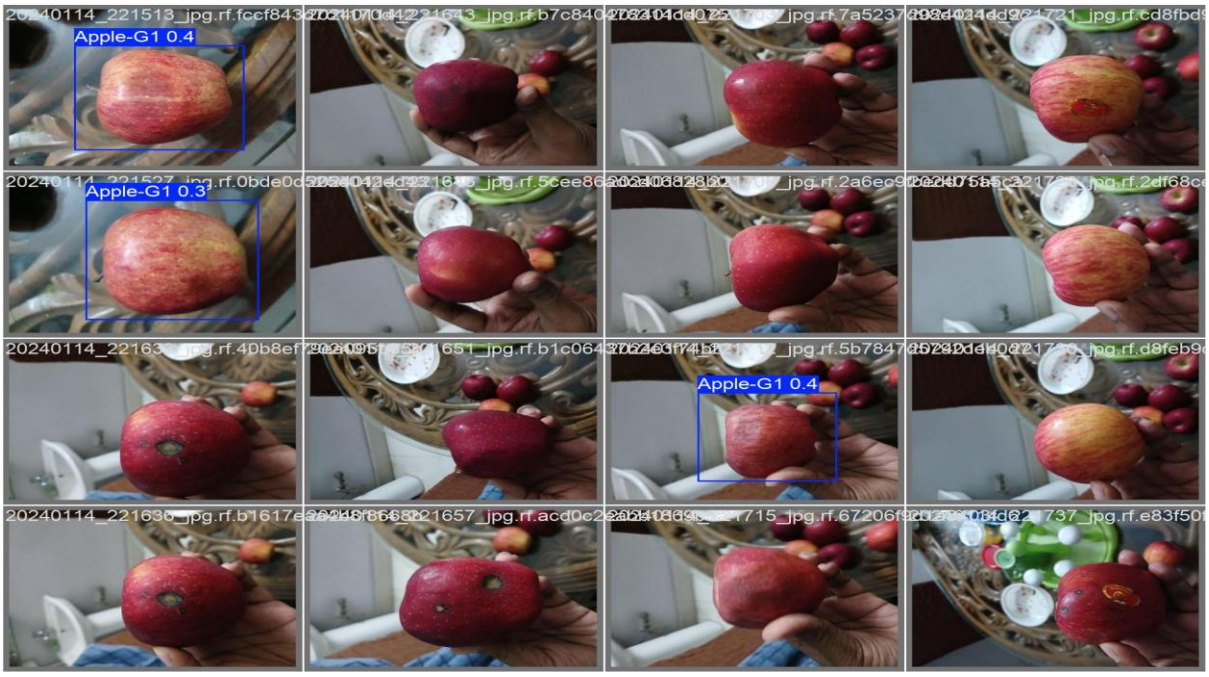
Trained model and results saved at:

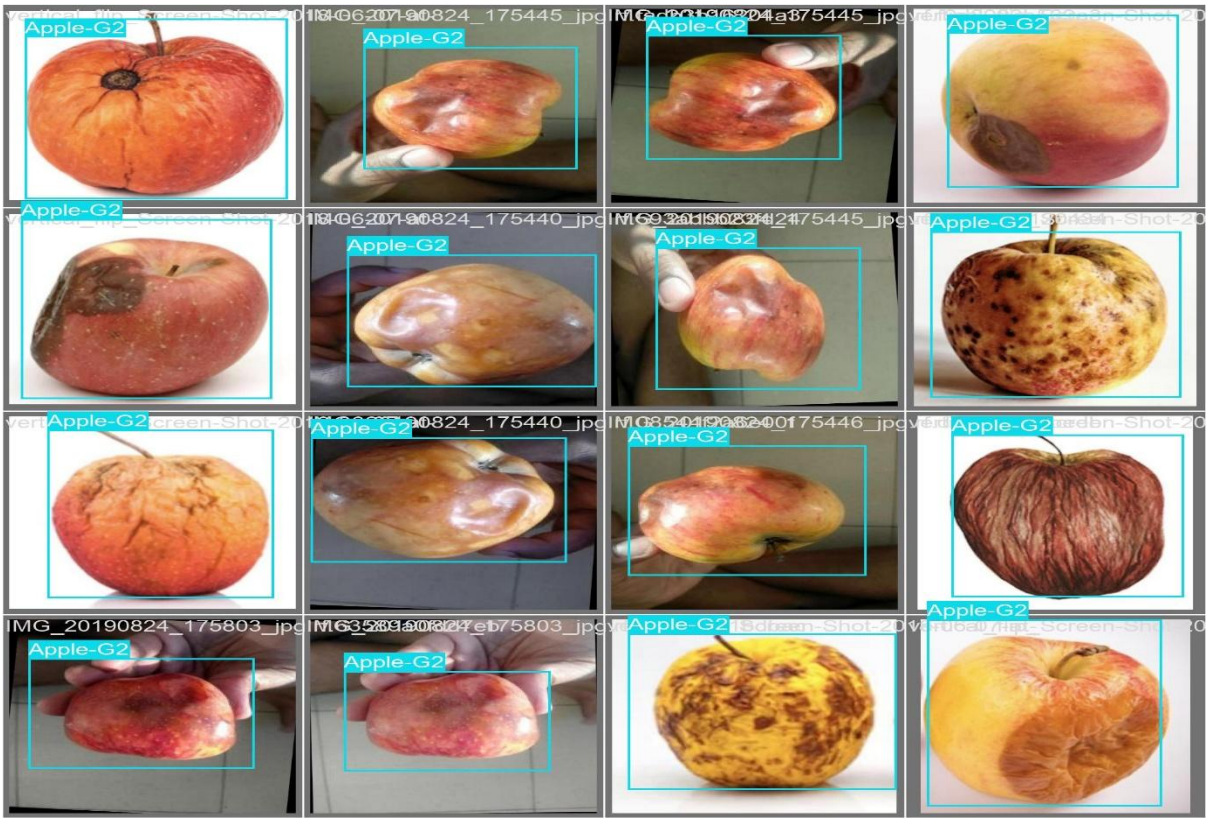
C:\Users\rekha\runs\detect\train-6

11. Sample Output



Apple-G1


Apple-G1 0.4
Apple-G1 0.3
Apple-G1 0.4


Apple-G2
Apple-G2
Apple-G2
Apple-G2
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Apple-G2
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Apple-G2
Apple-G2
Apple-G2
Apple-G2

Part B: Classification

- Train YOLO classification model
- Dataset: Roboflow classification dataset

Screen recording

https://drive.google.com/file/d/1QU7nPT8-XmdlKV72TQMTgJxXWwEh4P5-/view?usp=sharing

1. Title

Image Classification using YOLOv8-cls (Ultralytics Framework)

2. Objective

To design and implement an image classification system using YOLOv8-cls model to classify different types of flowers from a custom dataset with high accuracy.

3. Dataset Description

- **Source:** Roboflow Universe (Flower Dataset)
- **Dataset Link:** https://universe.roboflow.com/ambos-vgbi4/flower-dataset-lnsyp/dataset/1
- **Classes:** [Your flower classes - e.g., Daisy, Dandelion, Rose, Sunflower, Tulip]

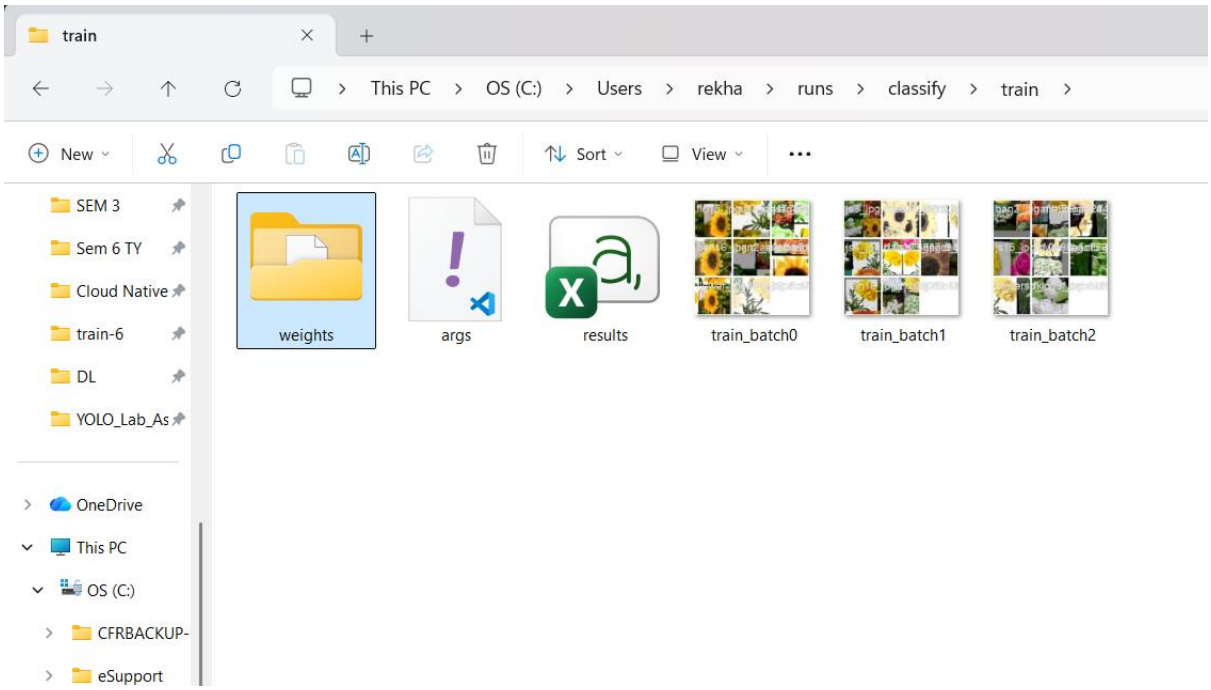
4. Training Command Used

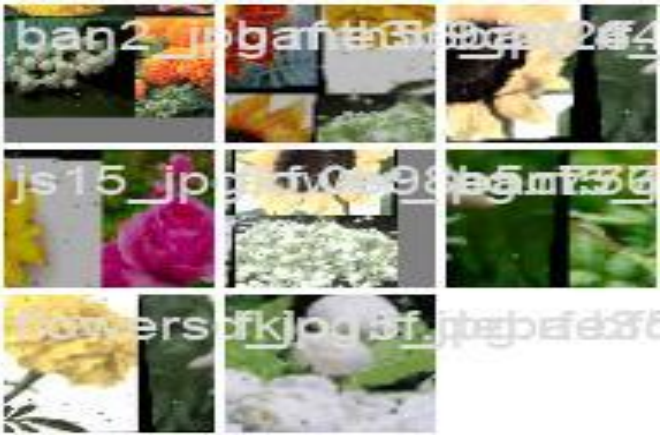
✓ yolo classify train model=yolov8n-cls.pt data=classification/ epochs=3 imgsz=64 batch=8

```
C:\Windows\System32\cmd.e
Starting training for 3 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
        1/3         0G          0      0.621          0          0         6
        1/3         0G          0     0.6218          0          0         6
        1/3         0G          0     0.6198          0          0         6
        1/3         0G          0     0.6191          0          0         6
        1/3         0G          0     0.6186          0          0         6
        1/3         0G          0     0.6174          0          0         6
        1/3         0G          0     0.6164          0          0         6
        1/3         0G          0     0.6152          0          0         6
        1/3         0G          0     0.6139          0          0         6
        1/3         0G          0     0.6129          0          0         6
        1/3         0G          0     0.6121          0          0         6
        1/3         0G          0     0.6116          0          0         6
        1/3         0G          0     0.6108          0          0         6
        1/3         0G          0     0.6101          0          0         6
        1/3         0G          0     0.6092          0          0         6
        1/3         0G          0     0.6085          0          0         6
        1/3         0G          0     0.6078          0          0         6
        1/3         0G          0      0.607          0          0         6
        1/3         0G          0     0.6062          0          0         6
        1/3         0G          0     0.6054          0          0         6
        1/3         0G          0     0.6045          0          0         6
        1/3         0G          0     0.6036          0          0         6
        1/3         0G          0     0.6027          0          0         6
        1/3         0G          0     0.5926          0          0         6
        1/3         0G          0     0.5926          0          0         6
4: 100% ━━━━━━━━━━━━ 24/24 4.0it/s 6.0s
                 Class     Images  Instances       Box(P          R      mAP5
                 Class     Images  Instances       Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 7.4it/s 0.3s
C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\venv\Lib\site-packa
ges\ultralytics\utils\metrics.py:839: RuntimeWarning: Mean of empty slice
  i = smooth(f1_curve.mean(0), 0.1).argmax()  # max F1 index
C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\venv\Lib\site-packa
ges\numpy\_core\_methods.py:134: RuntimeWarning: invalid value encountered i
n divide
  ret = um.true_divide(
                 all         19          0          0          0
0          0
WARNING no labels found in detect set, cannot compute metrics without labels

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Siz
e
```

```
C:\Windows\System32\cmd.e

        3/3         0G          0     0.5418          0          0          6
        3/3         0G          0     0.5418          0          0          6
        3/3         0G          0     0.5412          0          0          6
        3/3         0G          0     0.5407          0          0          6
        3/3         0G          0     0.5402          0          0          6
        3/3         0G          0     0.5399          0          0          6
        3/3         0G          0     0.5395          0          0          6
        3/3         0G          0     0.5389          0          0          6
        3/3         0G          0     0.5384          0          0          6
        3/3         0G          0      0.538          0          0          6
        3/3         0G          0     0.5375          0          0          6
        3/3         0G          0      0.537          0          0          6
        3/3         0G          0     0.5364          0          0          6
        3/3         0G          0     0.5359          0          0          6
        3/3         0G          0     0.5355          0          0          6
        3/3         0G          0     0.5351          0          0          6
        3/3         0G          0     0.5347          0          0          6
        3/3         0G          0     0.5343          0          0          6
        3/3         0G          0     0.5257          0          0          6
        3/3         0G          0     0.5257          0          0          6
4: 100% ━━━━━━━━━━━━━━━━ 24/24 5.4it/s 4.4s
                 Class     Images  Instances      Box(P          R      mAP5
                 Class     Images  Instances      Box(P          R      mAP5
0  mAP50-95): 100% ━━━━━━━━━━━━━━━━ 2/2 10.1it/s 0.2s
C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\venv\Lib\site-packa
ges\ultralytics\utils\metrics.py:839: RuntimeWarning: Mean of empty slice
  i = smooth(f1_curve.mean(0), 0.1).argmax()  # max F1 index
C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\venv\Lib\site-packa
ges\numpy\_core\_methods.py:134: RuntimeWarning: invalid value encountered i
n divide
  ret = um.true_divide(
                   all         19          0          0          0
0          0
WARNING no labels found in detect set, cannot compute metrics without labels

3 epochs completed in 0.005 hours.
Optimizer stripped from C:\Users\rekha\runs\classify\train\weights\last.pt,
6.2MB
Optimizer stripped from C:\Users\rekha\runs\classify\train\weights\best.pt,
6.2MB

Validating C:\Users\rekha\runs\classify\train\weights\best.pt...
Ultralytics 8.4.40  Python-3.11.9 torch-2.11.0+cpu CPU (AMD Ryzen 5 5600H wi
th Radeon Graphics)
Model summary (fused): 73 layers, 3,006,428 parameters, 0 gradients, 8.1 GFL
OPs
```

Folder


train
This PC
OS (C:)
Users
rekha
runs
classify
train
New
Sort
View
SEM 3
Sem 6 TY
Cloud Native
train-6
DL
YOLO_Lab_As
OneDrive
This PC
OS (C:)
CFRBACKUP-
eSupport
weights
args
results
train_batch0
train_batch1
train_batch2



Part C: Pose Estimation

- Train YOLO pose model
- Dataset: Roboflow / COCO format

Screen recording

https://drive.google.com/file/d/1rWuPHJ8_Lp7osob_xPRUxabg0tY4FTvO/view?usp=sharing

1. Title

Human Pose Estimation using YOLOv8-pose (Ultralytics Framework)

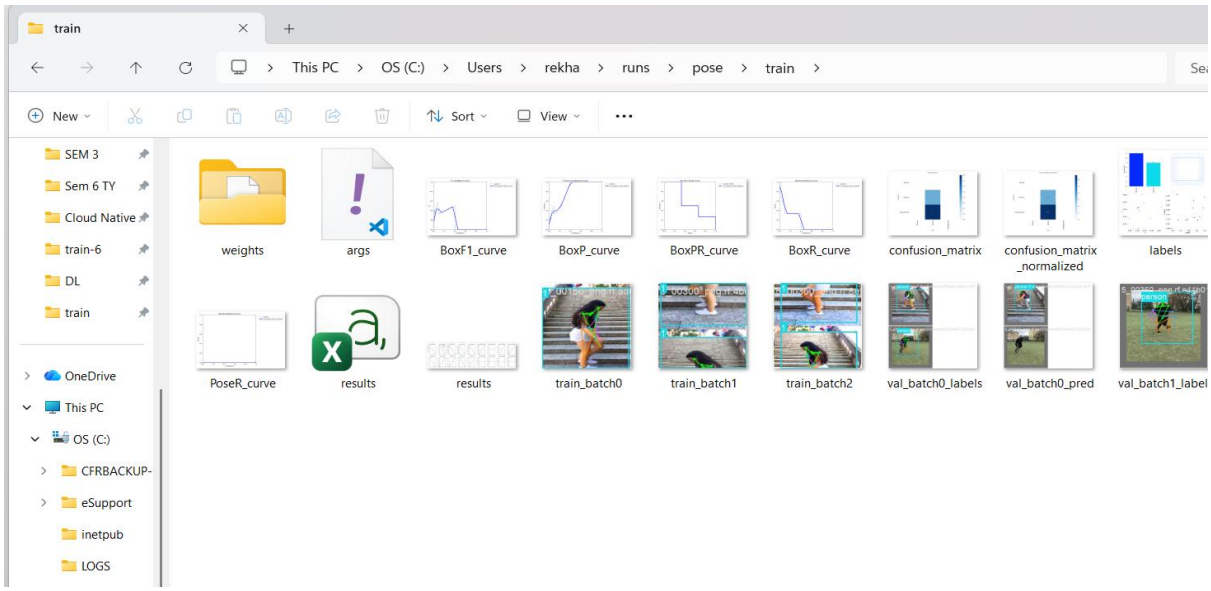
2. Objective

To design and implement a human pose estimation system using YOLOv8-pose model to detect and track human keypoints (body joints) from images.

3. Dataset Description

- **Source:** Roboflow Universe
- **Dataset Link:** https://universe.roboflow.com/viswanadh/pose-estimation-i4f6o/dataset/1
- **Format:** YOLOv8 Pose format

Folder

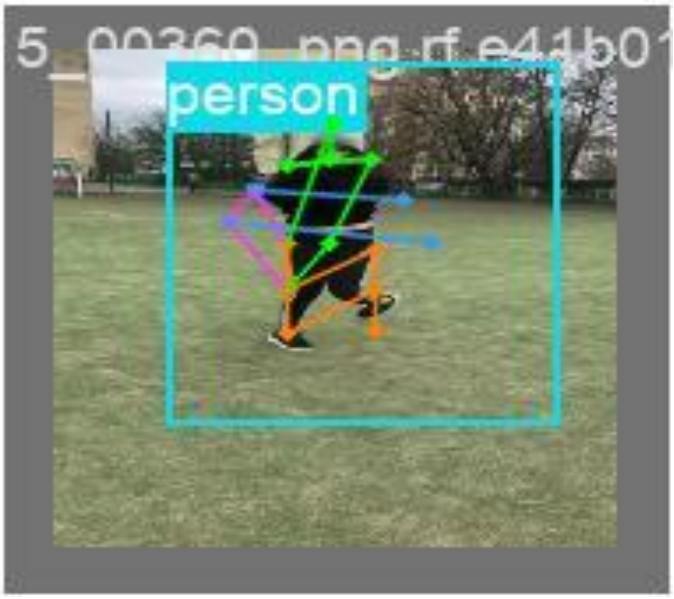


4. Training Command Used

yolo pose train model=yolov8n-pose.pt data=data.yaml epochs=2 imgsz=160 batch=1

```
C:\Windows\System32\cmd.e

      1/2         0G      2.667      9.417     0.7659      3.636      2.326          2
    160: 100% ━━━━━━━━━━━━ 12/12 3.6it/s 3.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
  Pose(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 9.5it/s 0.2s
                   all          3          3      0.758      0.333       0.41     0.0894
      0          0          0          0

     Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss   dfl_loss  Instances
   Size
      2/2         0G      2.654      10.39      1.217      3.817      2.231          4
      2/2         0G      2.569      6.939     0.8077      3.666      2.231          3
      2/2         0G      2.624      6.356     0.6997      3.669       2.24          2
      2/2         0G      2.754      5.929     0.6149      3.737      2.292          4
      2/2         0G      2.677      6.981     0.6004      3.678      2.283          2
      2/2         0G      2.721      7.471     0.6343      3.579      2.286          2
      2/2         0G      2.681      7.182     0.6023       3.54      2.288          2
      2/2         0G      2.518      7.599     0.6253      3.464      2.195          1
      2/2         0G      2.527      7.939     0.6962      3.605        2.2          4
      2/2         0G      2.562       8.05     0.7296      3.645      2.222          4
      2/2         0G      2.546      8.326      0.715      3.558      2.189          1
      2/2         0G      2.604      8.491     0.7532      3.499        2.2          1
      2/2         0G      2.604      8.491     0.7532      3.499        2.2          1
    160: 100% ━━━━━━━━━━━━ 12/12 6.9it/s 1.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
  Pose(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 14.0it/s 0.1s
                   all          3          3      0.859      0.333       0.59      0.141
      0          0          0          0

2 epochs completed in 0.002 hours.
Optimizer stripped from C:\Users\rekha\runs\pose\train\weights\last.pt, 6.8MB
Optimizer stripped from C:\Users\rekha\runs\pose\train\weights\best.pt, 6.8MB

Validating C:\Users\rekha\runs\pose\train\weights\best.pt...
Ultralytics 8.4.40  Python-3.11.9 torch-2.11.0+cpu CPU (AMD Ryzen 5 5600H with Radeon Graph
ics)
YOLOv8n-pose summary (fused): 82 layers, 3,290,159 parameters, 0 gradients, 9.2 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)
  Pose(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 2/2 11.2it/s 0.2s
                   all          3          3      0.859      0.333       0.59      0.141
      0          0          0          0
               person          3          3      0.859      0.333       0.59      0.141
      0          0          0          0
Speed: 0.5ms preprocess, 36.3ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to C:\Users\rekha\runs\pose\train
```

10. Sample Output


person

1
1

Deployment part

```
C:\Windows\System32\cmd.e

(venv) C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment>notepad deploy_pretrained.p
y

(venv) C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment>python deploy_pretrained.py

============================================================
MULTI-TASK VISION SYSTEM - DEPLOYMENT (Using Pretrained Models)
============================================================

✓ Webcam detected!

[1/3] Loading Object Detection Model...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
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Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt to 'y
olo11n.pt': 100% ━━━━━━━━━━━━ 5.4MB 3.4MB/s 1.6s
✓ Detection model ready

[2/3] Loading Classification Model...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
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Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
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Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-cls.pt t
o 'yolo11n-cls.pt': 100% ━━━━━━━━━━━━ 5.5MB 3.1MB/s 1.8s
✓ Classification model ready

[3/3] Loading Pose Estimation Model...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-pose.pt
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n-pose.pt
```
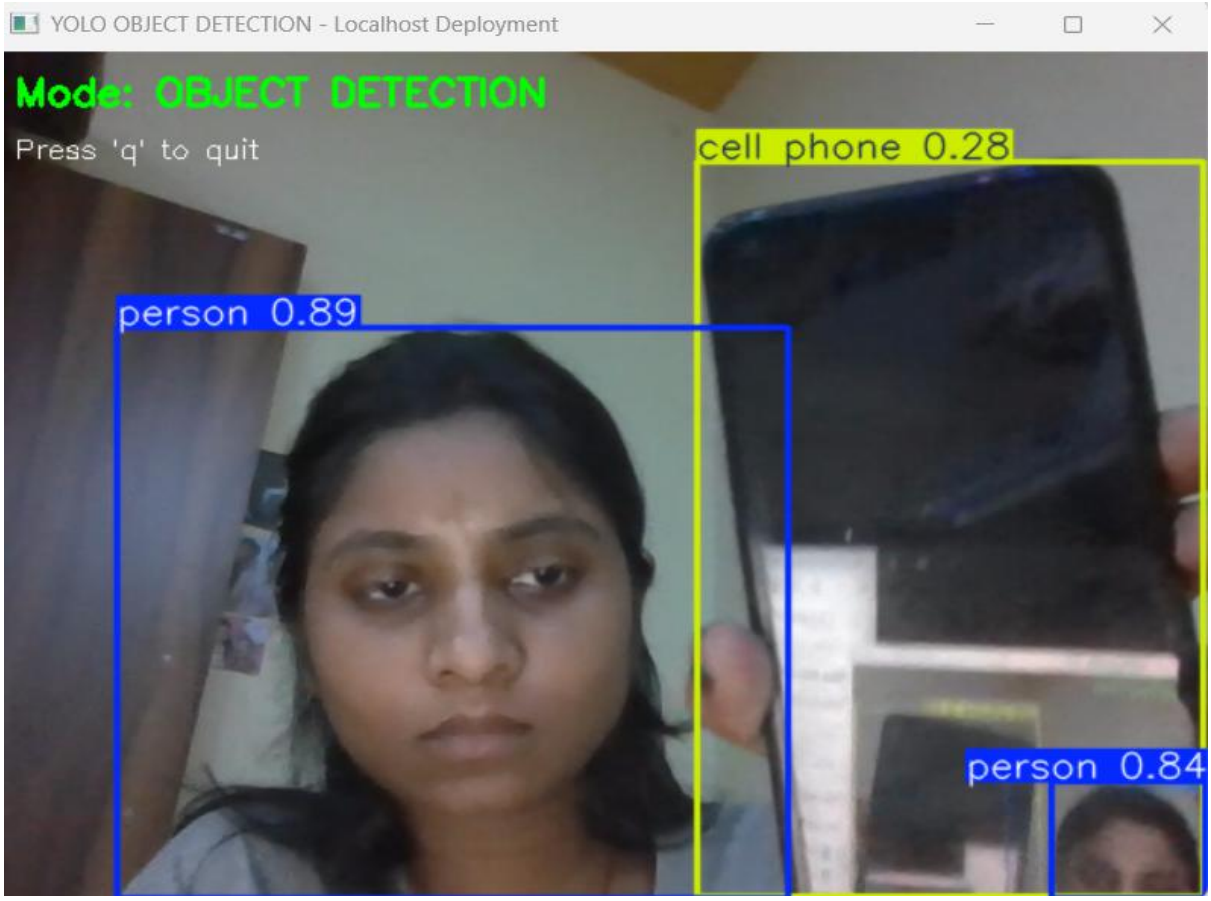
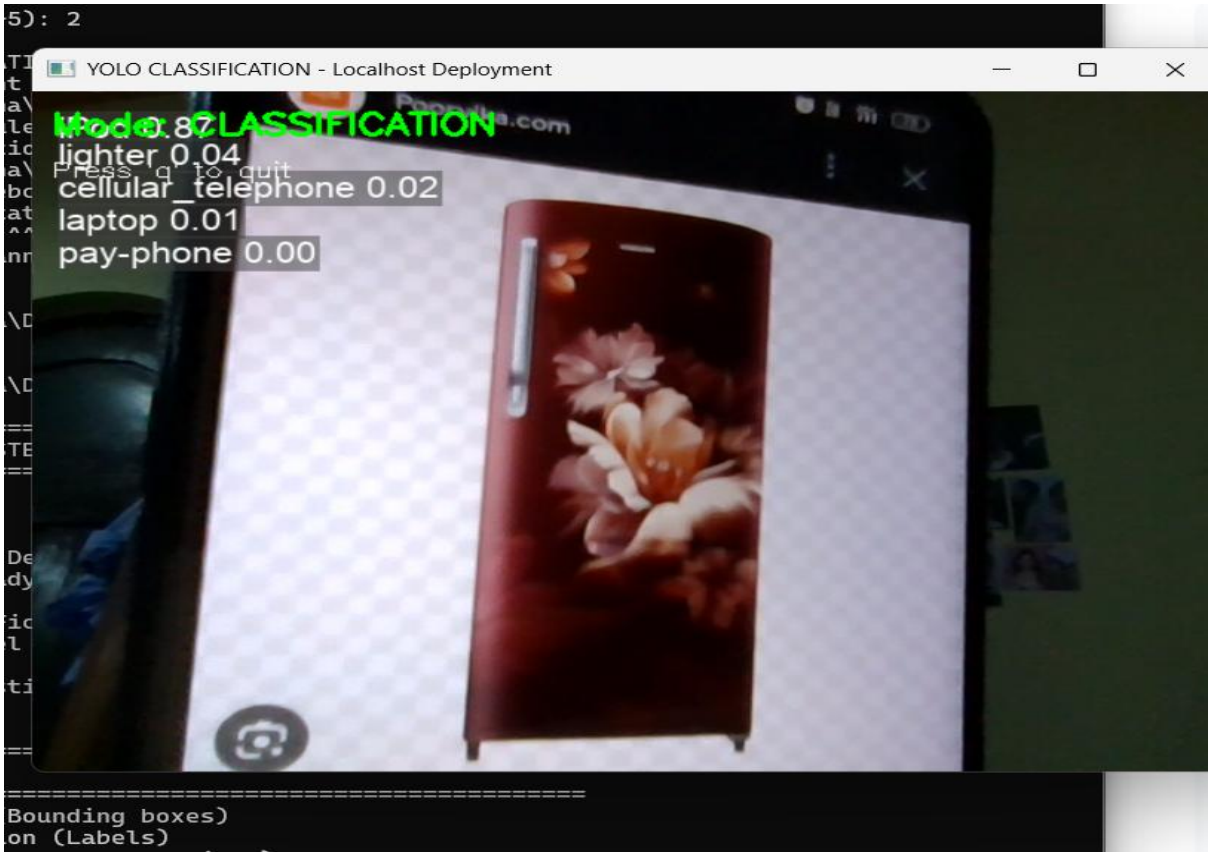
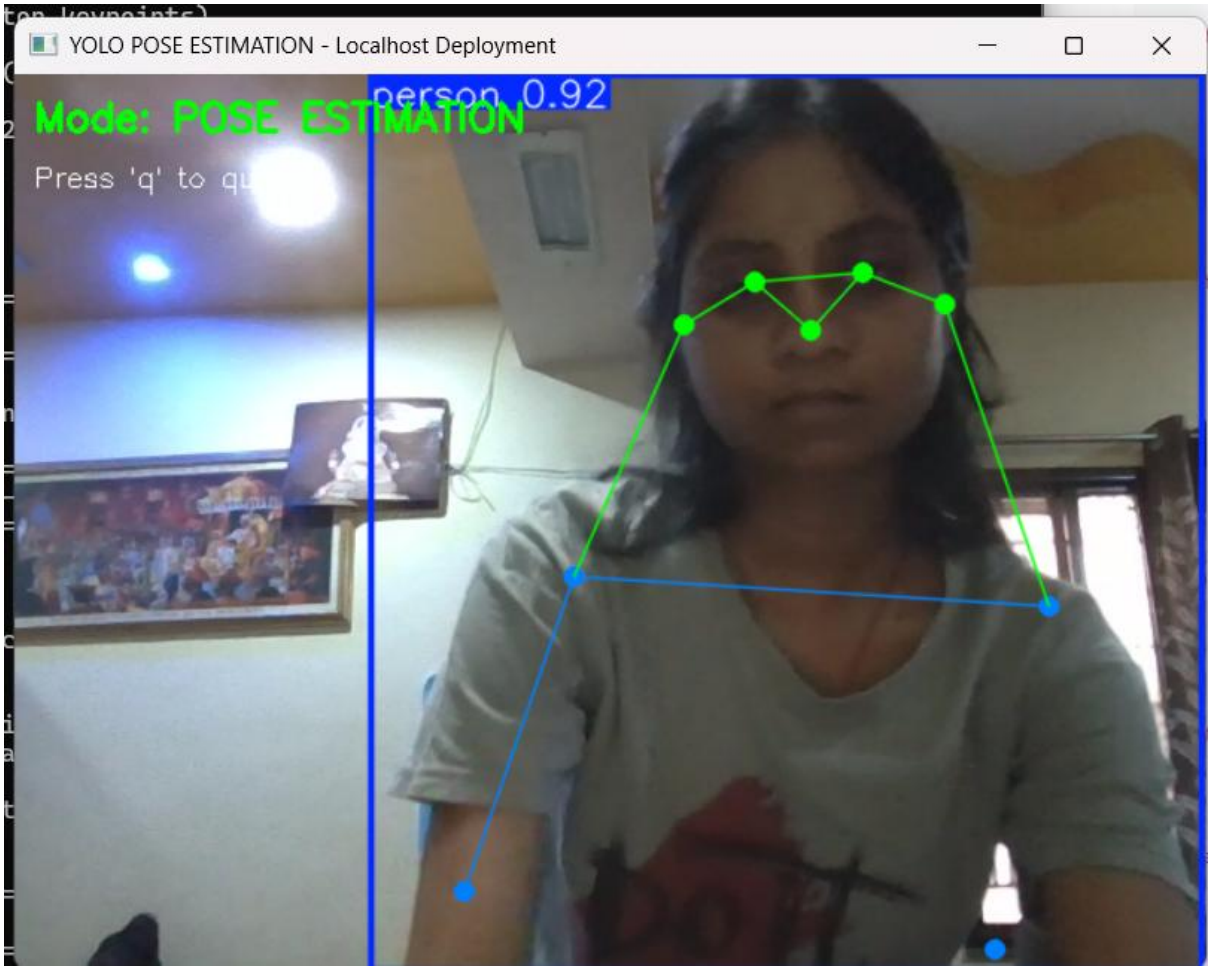
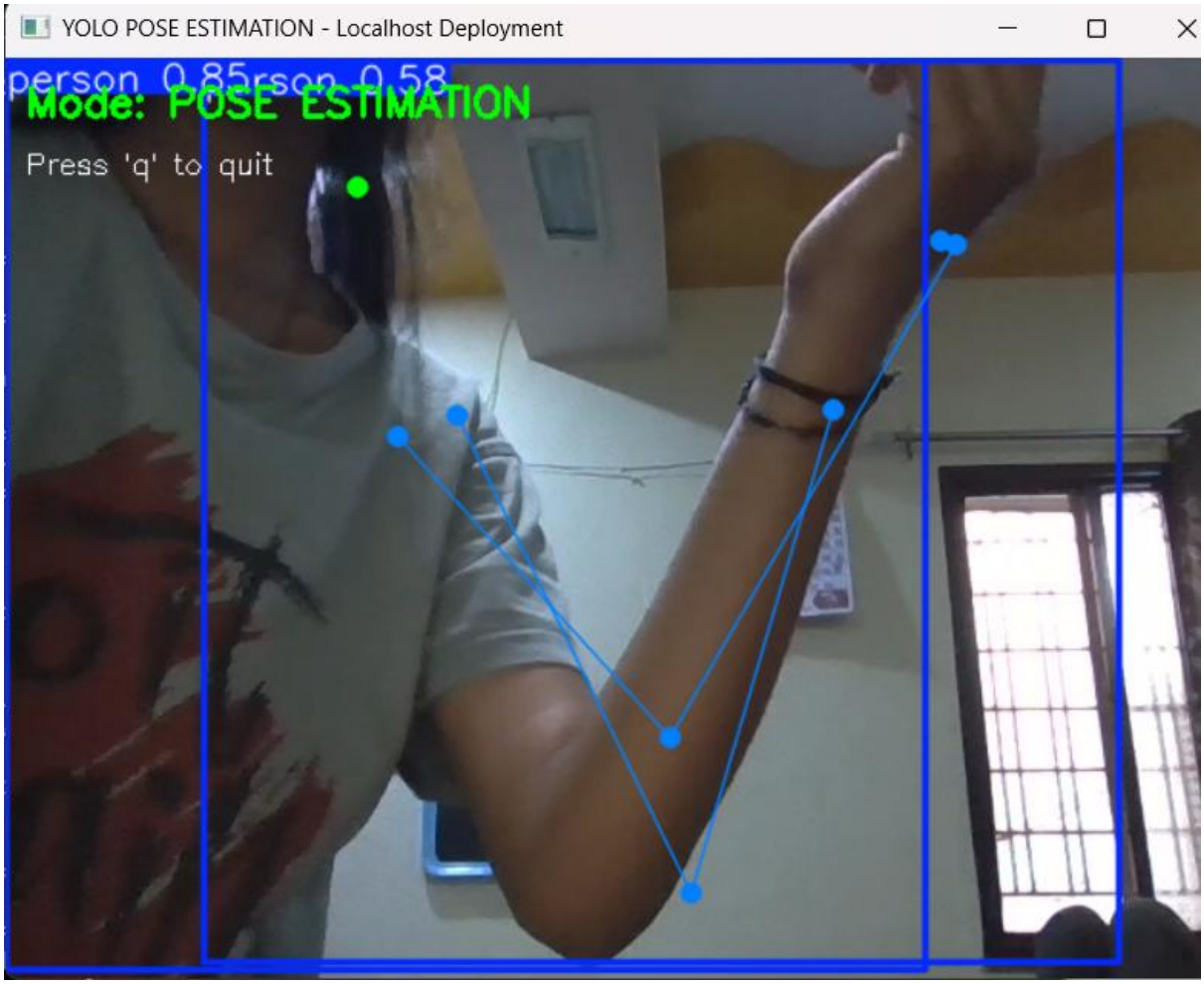

```
(venv) C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment>notepad deploy_pretrained_f
ixed.py

(venv) C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment>python deploy_pretrained_fi
xed.py
============================================================
MULTI-TASK VISION SYSTEM - DEPLOYMENT (Using Pretrained Models)
============================================================

✓ Webcam detected!

[1/3] Loading Object Detection Model...
✓ Detection model ready

[2/3] Loading Classification Model...
✓ Classification model ready

[3/3] Loading Pose Estimation Model...
✓ Pose model ready

============================================================
DEPLOYMENT MENU
============================================================
1. Object Detection (Bounding boxes)
2. Image Classification (Labels)
3. Pose Estimation (Skeleton keypoints)
4. ALL THREE sequentially
5. Test on sample images (No webcam needed)
```

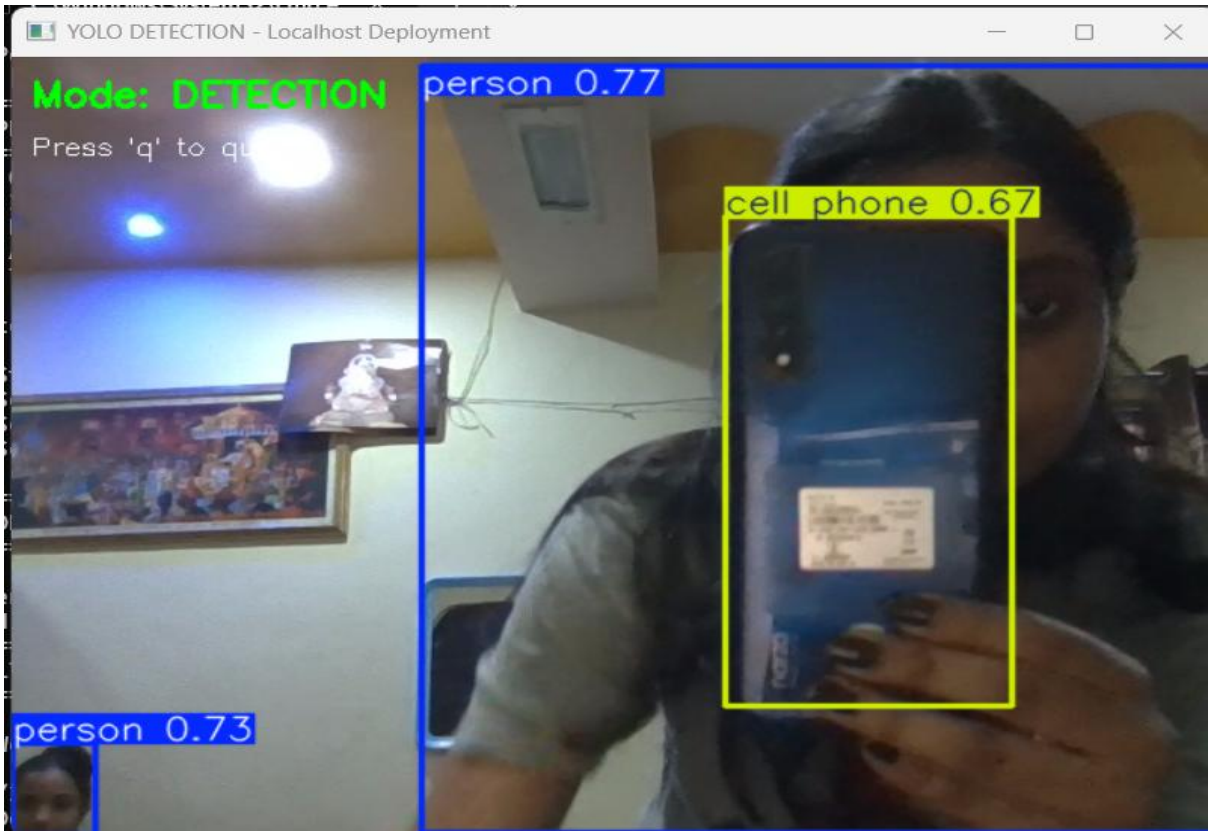


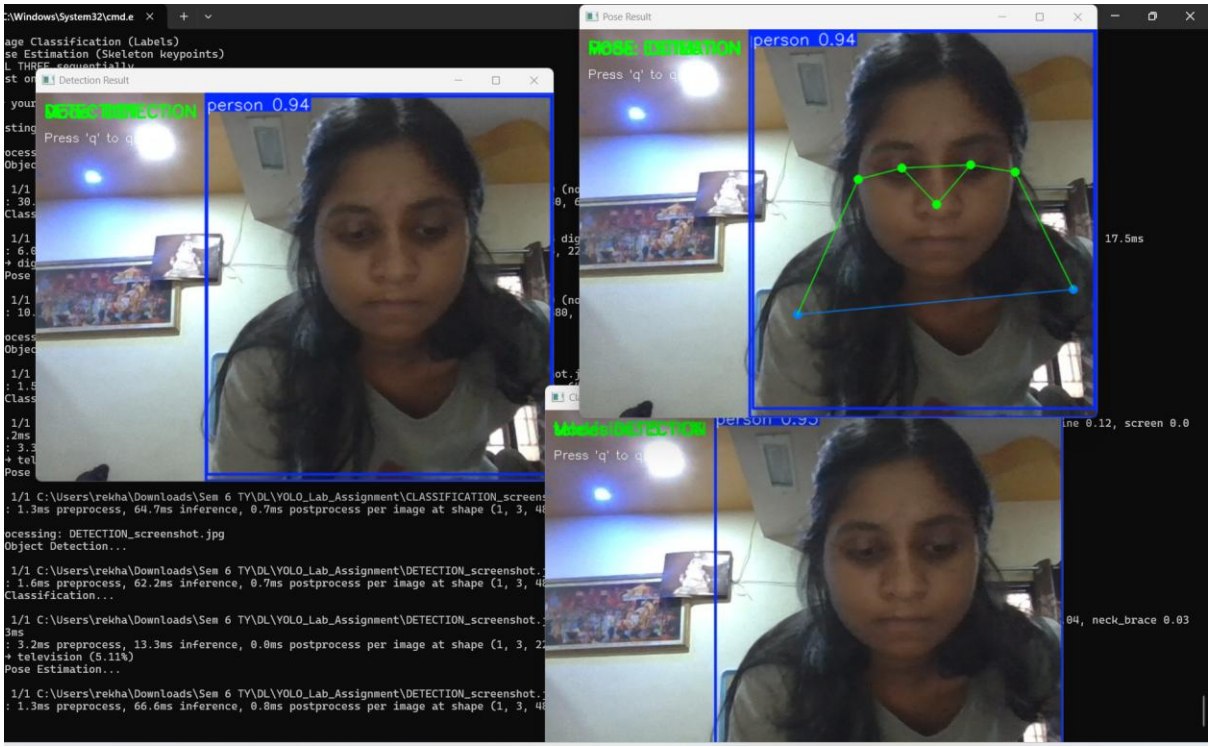
YOLO OBJECT DETECTION - Localhost Deployment
Mode: OBJECT DETECTION
Press 'q' to quit
toothbrush 0.79
cell phone 0.92
person 0.82
tion (Labels)
(Skeleton keypoints)


5): 2
YOLO CLASSIFICATION - Localhost Deployment
Mode: CLASSIFICATION
Press 'q' to quit
lighter 0.04
cellular_telephone 0.02
laptop 0.01
pay-phone 0.00
Bounding boxes)
on (Labels)


YOLO POSE ESTIMATION - Localhost Deployment
Mode: POSE ESTIMATION
Press 'q' to qu
person 0.92


YOLO POSE ESTIMATION - Localhost Deployment
Mode: POSE ESTIMATION
Press 'q' to quit


YOLO DETECTION - Localhost Deployment
Mode: DETECTION
person 0.77
cell phone 0.67
person 0.73




```
🧪 Testing on sample images...

📷 Processing: test_image.jpg
   🔍 Object Detection...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\test_image.jpg: 480x640 (no detections), 85.4ms
Speed: 30.0ms preprocess, 85.4ms inference, 0.7ms postprocess per image at shape (1, 3, 480, 640)
   🏷️ Classification...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\test_image.jpg: 224x224 digital_clock 0.11, nematode 0.06, web_site 0
Speed: 6.0ms preprocess, 17.5ms inference, 0.0ms postprocess per image at shape (1, 3, 224, 224)
     → digital_clock (11.04%)
   🕺 Pose Estimation...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\test_image.jpg: 480x640 (no detections), 114.8ms
Speed: 10.8ms preprocess, 114.8ms inference, 0.7ms postprocess per image at shape (1, 3, 480, 640)

📷 Processing: CLASSIFICATION_screenshot.jpg
   🔍 Object Detection...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\CLASSIFICATION_screenshot.jpg: 480x640 1 person, 57.3ms
Speed: 1.5ms preprocess, 57.3ms inference, 1.0ms postprocess per image at shape (1, 3, 480, 640)
   🏷️ Classification...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\CLASSIFICATION_screenshot.jpg: 224x224 television 0.20, cellular_tele
8, 11.2ms
Speed: 3.3ms preprocess, 11.2ms inference, 0.1ms postprocess per image at shape (1, 3, 224, 224)
     → television (20.50%)
   🕺 Pose Estimation...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\CLASSIFICATION_screenshot.jpg: 480x640 1 person, 64.7ms
Speed: 1.3ms preprocess, 64.7ms inference, 0.7ms postprocess per image at shape (1, 3, 480, 640)

📷 Processing: DETECTION_screenshot.jpg
   🔍 Object Detection...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\DETECTION_screenshot.jpg: 480x640 1 person, 62.2ms
Speed: 1.6ms preprocess, 62.2ms inference, 0.7ms postprocess per image at shape (1, 3, 480, 640)
   🏷️ Classification...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\DETECTION_screenshot.jpg: 224x224 television 0.05, cellular_telephone
, 13.3ms
Speed: 3.2ms preprocess, 13.3ms inference, 0.0ms postprocess per image at shape (1, 3, 224, 224)
     → television (5.11%)
   🕺 Pose Estimation...

image 1/1 C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment\DETECTION_screenshot.jpg: 480x640 1 person, 66.6ms
Speed: 1.3ms preprocess, 66.6ms inference, 0.8ms postprocess per image at shape (1, 3, 480, 640)

✓ Testing complete!

============================================================
✓ DEPLOYMENT COMPLETE
============================================================

(venv) C:\Users\rekha\Downloads\Sem 6 TY\DL\YOLO_Lab_Assignment>
```